Prasad Sadanand Mahabare

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TECHNICAL SKILLS

- **Programming: Embedded C, Python, Bash**
- **Microcontrollers: PIC18F4520, STM32F446RE, ESP32-WROOM**
- **Developer Tools: VsCode, Keil, MPLabx, STM32CubeIDE, ArduinoIDE**
- **Simulation Platforms: Proteus8, PicSimLab, WokWi**
- **Protocols: UART, I2C, SPI, MQTT**
- **IoT Cloud Platform: Blynk, ThinkSpeak, RedNode**

PROFESSIONAL EXPERIENCE

Student Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Configured and installed Cisco catalysts (2650s to 9300s), firewalls, load balancers, DNS.
- Maintained Purdue Cloud System, performance by **applying hot fixes** and **security configurations.**
- Developed scripts to **automate firmware upgrades** over 800 network devices.

PROJECTS

ESP32 OTA Firmware Management System | Github

- Developed **WiFi-enabled ESP32** software for remote **OTA upgrades**, **watchdog-based recovery**, and live device telemetry.
- Integrated **FreeRTOS** task orchestration with **MQTT** messaging for deployment pipelines and remote maintenance.
- Designed a **cloud-linked dashboard** supporting firmware version visibility, update triggering, and node status tracking.

STM32 FreeRTOS System Health Monitor | Github

- Created a multitasking **STM32 Cortex-M4** application using **FreeRTOS** for deterministic scheduling and runtime supervision.
- Integrated **UART diagnostics**, **ADC sampling**, heartbeat indication, and anomaly detection through concurrent execution flows.
- Configured **GPIO, ADC, UART,** and timer peripherals in **STM32CubeIDE** using priority-driven embedded architecture.

PIC18F4520 UART Command & Peripheral Management System | Github

- Built a **UART-controlled** management interface on **PIC18F4520** for command-driven operation of embedded peripherals.
- Designed **interrupt-based serial handling** and parser logic for stable low-latency communication.
- Performed hardware validation through **Proteus simulation**, peripheral testing, and **serial trace debugging**.

EDUCATION

P.G Diploma in Embedded Systems Design **Sep 2025 - May 2026**
BICARD Institute, Pune

Master of Science in Computer Science **Jan 2022 - Dec 2023**
Purdue University, Fort Wayne, Indiana

Bachelor of Engineering Computer Engineering **Aug 2016 - Aug 2019**
Pune University, Pune, India